SECTION C: BINDING TO GLOBAL STATE 1004

```
GlobalState (10-Field Poseidon Commitment)

globalStateRoot = Poseidon.hash([
   slot 0: walletRegistryRoot
   slot 1: walletCount
   slot 2: adminPublicKeyHash
   slot 3: lifecycleRoot
   slot 4: benefitPlanRoot
   slot 5: credentialRegistryRoot
   slot 6: eligibilityVkHash
   slot 7: messagingRoot
   slot 8: auditChainHash <-- AUDIT HEAD
   slot 9: blockHeight
])
```

```
    Binding Chain:

auditChainHash (latest chainHash)
            |
            v
GlobalState (10-field Poseidon commitment)
            |
            v
ChainProof (recursive ZK proof, ~22KB)

Single proof encapsulates: wallet registry,
lifecycle, benefit plans, credentials,
messaging, ALL audit trail history,
and block height.
```

SECTION D: SEVEN-CHECK AUDIT SUITE 1006

```
AuditService.runFullAudit() -- stateless, no persistent state
Input: AuditLog, L2Batch[], OnChainState, rosterEntries[]
Output: FullAuditReport { overallHealthy, 7 check results }
```

#	CHECK NAME	SEVERITY	WHAT IT VERIFIES
1	Chain Integrity	CRITICAL	Walk from genesis, recompute each chainHash. Poseidon.hash([prev, commit]) == event.chainHash
2	Financial Balance	CRITICAL	patientPay + planPay == ingredientCost + fee for every claim event with claimContext
3	Duplicate Detection	CRITICAL	Group by walletId | claimContext.ndcHash | claimContext.dateOfService. Multiple receipts in same group = duplicate
4	Sequence Gaps	CRITICAL	Each event.seqNum == previous.seqNum + 1. Any gap = missing event (data loss/corruption)
5	Orphaned Receipts	INFO	Events with status=='pending' older than 10 min. Expected during normal batch intervals
6	Settlement Verification	ROOT-CRIT	Latest settled batch roots must match on-chain state (registryRoot, lifecycleRoot)
7	Enrollment Consistency	CRITICAL	L2 registration events vs roster entries. Detect unregistered or unrostered wallets

OVERALL HEALTH FORMULA:

```
overallHealthy = chainIntegrity.isValid && financialBalance.isValid
   && !duplicates.hasDuplicates && !sequenceGaps.hasGaps
   && enrollment.isConsistent && settlement.rootConsistent
EXCLUDED: orphanedReceipts (pending work is normal), incomplete settlement (expected)
```